

Shile Wang

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Education

Peking University (PKU), Beijing, China

09/2023 – Present

Bachelor of Mathematics and Applied Mathematics, School of Mathematical Sciences

- Major GPA: 89/100 (3.72/4.00)
- Related Courses: Geometry; Differential Geometry; Topology; Differential Manifold; Homology Theory; Geometric Group Theory; Characteristic Classes
- Research Interests: Low dimensional topology; Geometric group theory

Research Experience

Fixed Point Properties of Group Actions on CAT(0) Spaces

06/2026 – 08/2026

University of Chicago Mathematics REU

Mentor: Zhong Zhang

- Studied J.-P. Serre's *Trees*, B. Farb's *Group Actions and Helly's Theorem*, and M. R. Bridson's *Semisimple Actions of Mapping Class Groups on CAT(0) Spaces*.
- Learned about how to prove property FA_n for reflection groups, arithmetic groups, mapping class groups, and braid groups.
- Wrote an expository article: [*Fixed Point Properties of Group Actions on CAT\(0\) Spaces*](#).
- Gave a talk on fixed point properties of group actions on trees.

Mapping Class Group

09/2025 – 06/2026

Instructor: Prof. Yi Liu (Beijing International Center for Mathematical Research)

- Studied B. Farb and D. Margalit's *A Primer on Mapping Class Groups*.
- Became familiar with the basic tools used to study mapping class groups, such as the Alexander method, properties of Dehn twists, the Birman exact sequence, the symplectic representation, and the Johnson homomorphism.
- Learned the structure of mapping class groups via the Dehn-Nielsen-Baer theorem, and studied basic properties of Teichmüller space such as Fenchel-Nielsen coordinates and Teichmüller's theorems on Teichmüller's extremal problem.

CAT(0) Cube Complexes

03/2026 – 06/2026

Peking University Low-Dimensional Topology Seminar

Instructors: Prof. Yi Liu, Prof. Jiajun Wang, Prof. Wenyuan Yang

- Studied P. Schwer's *CAT(0) Cube Complexes: An Introduction*.
- Learned the basic properties of CAT(0) cube complexes and studied cubulations of half-space systems.
- Presented part of J. Crisp and B. Wiest's *Quasi-isometrically embedded subgroups of braid and diffeomorphism groups*, focusing on how to embed right-angled Artin groups into mapping class groups.

Knots and Links

10/2025 – 12/2025

Peking University Low-Dimensional Topology Seminar

Instructors: Prof. Yi Liu and Prof. Jiajun Wang

- Studied D. Rolfsen's *Knots and Links* and presented part of Chapter 3.
- Learned the Wirtinger presentation and used it to compute the fundamental groups of knots and links and to study the properties of horned spheres.
- Learned basic techniques to study knots and links, such as Seifert surfaces and the Alexander invariant, which is a powerful invariant to distinguish knots.

CAT(0) Spaces

09/2025 – 12/2025

Peking University Geometric Group Theory Seminar

Instructor: Prof. Wenyuan Yang

- Studied Sections 2.1–2.8 of M. R. Bridson and A. Haefliger's *Metric Spaces of Non-Positive Curvature* and presented Section 2.6.
- Learned the definition and basic properties of CAT(0) spaces, such as convexity and limits in CAT(0) spaces.
- Learned to prove the Cartan-Hadamard theorem using the exponential map and Alexandrov's Patchwork.
- Studied the general structure of groups of isometries of CAT(0) spaces and the properties of Clifford translations; also became familiar with cone topology, horofunctions, and Busemann functions.

Heegaard Splitting

03/2025 – 06/2025

Peking University Low-Dimensional Topology Seminar

Instructors: Prof. Yi Liu and Prof. Jiajun Wang

- Studied M. Scharlemann's *Heegaard splittings of compact 3-manifolds* and presented Section 2.
- Learned the definition of Heegaard splitting and Heegaard diagram, and how to read off basic algebraic topological information from a Heegaard diagram.
- Became familiar with basic tools in low dimensional topology such as handle decomposition and Morse theory.

Geometry of Surfaces

02/2025 – 06/2025

Peking University Geometry Seminar

Instructor: Assoc. Prof. Zhiqiang Bao

- Studied J. Stillwell's *Geometry of Surfaces* and presented Sections 5.6–6.10.
- Learned the classification of Euclidean, spherical, and hyperbolic surfaces (Killing-Hopf theorem) by studying the groups of isometries of surfaces.
- Studied the classification of closed geodesic paths by finding the relations between them and the fundamental groups of surfaces.
- Learned basic results in tessellation such as the properties of fundamental regions and orbifolds.

Honors & Awards

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| • Second Prize, Chinese College Student Mathematical Contest in Beijing area | 2024 |
| • “Yizheng” Scholarship | 2023 |
| • Second Prize, Chinese College Student Mathematical Contest in Beijing area | 2023 |
| • Gold Medal, Chinese Mathematical Olympiad (CMO) | 2022 |
| • Gold Medal, Chinese Girls' Mathematical Olympiad (CGMO) | 2022 |

Other Information

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- Lifelong passion and solid background for mathematics: participated in Mathematical Olympiad since primary school and won many awards.

- Standardized tests: GRE Mathematics Subject Test: 970 (95%).
- Strengths and interests: enjoy swimming (won three silver medals and one bronze medal at PKU) and volunteering.